

Richard Li

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Employment

- 2025 - Present **University of Washington** - *Postdoctoral Fellow*
Mentors: James Fogarty (DUB Group), Shwetak Patel (UbiComp Lab), and Cindy Lin (School of Medicine)
- 2019 - 2025 **University of Washington** - *Graduate Research Assistant*
Mentors: James Fogarty (DUB Group) and Shwetak Patel (UbiComp Lab)
- 2022 **Microsoft Research** - *Research Intern*
Mentors: Teddy Seyed, Nicolai Marquardt, Nathalie Riche, and Ken Hinckley (EPIC Team)
- 2021 **Microsoft Research** - *Research Intern*
Mentors: Eyal Ofek, Nicolai Marquardt, and Ken Hinckley (EPIC Team)
- 2020 **Microsoft Research** - *Research Intern*
Mentors: Alex Fiannaca and Ed Cutrell (Ability Team)
- 2019 - 2020 **X, the Moonshot Factory (Google [x])** - *Research Scholar*
Mentors: JB Berent and Yash Ahluwalia
- 2018 - 2019 **Google Research** - *Research Scholar*
Mentors: Dimitri Kanvesky, Thad Starner, Dick Lyon
- 2017 - 2019 **Georgia Institute of Technology** - *Graduate Research Assistant*
Mentors: Thad Starner and Gregory Abowd
- 2017 **Nvidia Research** - *Research Intern*
Mentor: Kaan Akşit
- 2015 **Amazon.com** - *Software Engineering Intern*
- 2014 **Intel** - *Research and Development Intern*
- 2013 **Broadcom** - *Software Configuration Management Intern*

Education

- 2019 - 2025 **University of Washington**
Ph.D. in Computer Science & Engineering
Advisors: James Fogarty and Shwetak Patel
- 2016 - 2018 **Georgia Institute of Technology**
M.S. in Computer Science
Advisors: Gregory Abowd and Thad Starner
- 2012 - 2016 **Georgia Institute of Technology**
B.S. in Computer Science, minor in Music Performance (clarinet)

Awards

- 2025 **Young Researcher** - *Heidelberg Laureate Forum*
- 2025 **CHI 2025 Best Paper Award** - *CHI Conference on Human Factors*
- 2024 **Travel Grant to UbiComp/ISWC 2024** - *UbiComp/ISWC*
- 2024 **I-Corps Participant** - *UW NSF Innovation Corps Site*

2024	Innovation Gap Fund (~\$10k) - UW CoMotion Labs
2023	Inventor of the Year Finalist - UW School of Medicine
2023	Husky 100 - University of Washington
2023	Population Health Pilot Grant (~\$100k) - UW Population Health Initiative
2019	Futuremaker Fellowship (~\$50k) - Siemens
2017	Runner-Up Master's Thesis - GT Three Minute Thesis Competition
2017	Outstanding Master's Thesis - GT College of Computing
2017	Best Computing Poster Award - GT Career, Research, and Innovation Development Conference
2015	Travel Grant to CSCW 2015 - GT GVU Center
2015	President's Undergraduate Research Award - GT Undergraduate Research Opportunities Program
2014 - 2016	HOPE Scholarship - GT Office of Scholarships and Financial Aid

Refereed Publications

2025	Beacon: At-Home, Self-Administration of a Test for Critical Flicker Frequency as a Screening Test for Minimal Hepatic Encephalopathy <i>AASLD The Liver Meeting</i> Philip Vutien*, <u>Richard Li</u> *, Ravi Karkar, James Fogarty, Sean A Munson, Sabrina Omer, Michael Yacoub, James Kashima, George N Ioannou 🗣️ Oral Presentation
2025	What Motivates Bladder Cancer Patients to be Active? A Qualitative Study Assessing Attitudes Towards Physical Activity and Digital Health Technologies <i>Urologic Oncology: Seminars and Original Investigations</i> Alex Zhu, Jessica Sher, <u>Richard Li</u> , Otari Ioseliani, Leah Cantor, Elena G Brewer, Karla Landis, David Bridges, Sean A Munson, Hanna Hunter, Cindy Lin, Sarah P Psutka
2025	Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency <i>ACM CHI Conference on Human Factors in Computing Systems</i> <u>Richard Li</u> , Philip Vutien, Sabrina Omer, Michael Yacoub, George Ioannou, Ravi Karkar, Sean A Munson, James Fogarty 🏆 Best Paper Award
2025	ECG Necklace: Low-power Wireless Necklace for Continuous ECG monitoring <i>ACM CHI Conference on Human Factors in Computing Systems</i> Qiuyue Xue, Eric Steven Martin, Jiaqing Liu, Ruiqing Wang, Antonio Glenn, <u>Richard Li</u> , Vikram Iyer, Shwetak Patel
2023	Evaluating a Novel, Portable, Self-Administrable Device ("Beacon") that Measures Critical Flicker Frequency as a Test for Hepatic Encephalopathy <i>ACG American Journal of Gastroenterology</i> Philip Vutien*, <u>Richard Li</u> *, Ravi Karkar, Sean A Munson, James Fogarty, Kara Walter, Michael Yacoub, George N Ioannou
2023	AdHocProx: Sensing Mobile, Ad-Hoc Collaborative Device Formations using Dual Ultra-Wideband Radios <i>ACM CHI Conference on Human Factors in Computing Systems</i> <u>Richard Li</u> , Teddy Seyed, Nicolai Marquardt, Eyal Ofek, Steve Hodges, Mike Sinclair, Hugo Romat, Michel Pahud, Jatin Sharma, William AS Buxton, Ken Hinckley, Nathalie Riche
2022	Food, Mood, Context: Examining College Students' Eating Context and Mental Well-Being

- ACM Transactions on Computing for Healthcare*
 Mehrab Bin Morshed, Samruddhi Shreeram Kulkarni, Koustuv Saha, [Richard Li](#), Leah G Roper, Lama Nachman, Hong Lu, Lucia Mirabella, Sanjeev Srivastava, Kaya De Barbaro, Munmun De Choudhury, Thomas Plötz, Gregory Abowd
- 2022 **SilentSpeller: Towards Mobile, Hands-Free, Silent Speech Text Entry using Electropalatography**
ACM CHI Conference on Human Factors in Computing Systems
 Naoki Kimura, Tan Gemicioglu, Jonathan Womack, [Richard Li](#), Yuhui Zhao, Abdelkareem Bedri, Zixiong Su, Alex Olwal, Jun Rekimoto, Thad Starner
- 2021 **Understanding the Design Space of Mouth Microgestures**
ACM Designing Interactive Systems Conference
 Victor Chen, Xuhai Xu, [Richard Li](#), Yuanchun Shi, Shwetak Patel, Yuntao Wang
- 2021 **Augmented Silkscreen: Designing AR Interactions for Debugging Printed Circuit Boards**
ACM Designing Interactive Systems Conference
 Ishan Chatterjee, Olga Khvan, Tadeusz Pforte, [Richard Li](#), Shwetak Patel
- 2020 **A Real-Time Eating Detection System for Capturing Eating Moments and Triggering Ecological Momentary Assessments to Obtain Further Context: System Development and Validation Study**
JMIR mHealth and uHealth
 Mehrab Bin Morshed, Samruddhi Shreeram Kulkarni, [Richard Li](#), Koustuv Saha, Leah Galante Roper, Lama Nachman, Hong Lu, Lucia Mirabella, Sanjeev Srivastava, Munmun De Choudhury, Kaya de Barbaro, Thomas Ploetz, Gregory D Abowd
- 2020 **Optical Gaze Tracking with Spatially-Sparse Single-Pixel Detectors**
IEEE International Symposium on Mixed and Augmented Reality
[Richard Li](#), Eric Whitmire, Michael Stengel, Ben Boudaoud, Jan Kautz, David Luebke, Shwetak Patel, Kaan Akşit
- 2020 **KissGlass: Greeting Gesture Recognition using Smart Glasses**
Augmented Humans International Conference
[Richard Li](#), Juyoung Lee, Woontack Woo, Thad Starner
- 2019 **Prediction of Mood Instability with Passive Sensing**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Mehrab Bin Morshed, Koustuv Saha, [Richard Li](#), Sidney K D'Mello, Munmun De Choudhury, Gregory D Abowd, Thomas Plötz
- 2019 **ScratchThat: Supporting Command-Agnostic Speech Repair in Voice-Driven Assistants**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Jason Wu, Karan Ahuja, [Richard Li](#), Victor Chen, Jeffrey Bigham
- 2019 **TongueBoard: An Oral Interface for Subtle Input**
Augmented Humans International Conference
[Richard Li](#), Jason Wu, Thad Starner
- 2018 **ScratchVR: Low-Cost, Calibration-Free Sensing for Tactile Input on Mobile Virtual Reality Enclosures**
ACM International Symposium on Wearable Computers
[Richard Li](#), Victor Chen, Gabriel Reyes, Thad Starner
- 2018 **Buccal: Low-Cost Cheek Sensing for Inferring Continuous Jaw Motion in Mobile Virtual Reality**
ACM International Symposium on Wearable Computers
[Richard Li](#), Gabriel Reyes

- 2018 **WristWash: Towards Automatic Handwashing Assessment Using a Wrist-Worn Device**
ACM International Symposium on Wearable Computers
 Hong Li, Shishir Chawla, [Richard Li](#), Sumeet Jain, Gregory D Abowd, Thad Starner, Cheng Zhang, Thomas Plötz
- 2017 **EarBit: Using Wearable Sensors to Detect Eating Episodes in Unconstrained Environments**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Abdelkareem Bedri, [Richard Li](#), Malcolm Haynes, Raj Prateek Kosaraju, Ishaan Grover, Temiloluwa Prioleau, Min Yan Beh, Mayank Goel, Thad Starner, Gregory Abowd

Lightly-Reviewed Publications

- 2025 **Towards Characterizing Knowledge Distillation of PPG Heart Rate Estimation Models**
NeurIPS Workshop on Learning from Time Series for Health
 Kanav Arora, Girish Narayanswamy, Shwetak Patel, [Richard Li](#)
- 2025 **Investigating the Use of Beacon for Cognitive Monitoring in People with Alzheimer's Disease and Lewy Body Dementia via the Critical Flicker Frequency**
AMIA Workgroup on Interactive Systems in Healthcare
[Richard Li](#), Philip Vutien, Ravi Karkar, Sabrina Omer, Michael Yacoub, Sarah Payne, Sean A Munson, James Fogarty, George Ioannou, Debby Tsuang
- 2024 **Improving Medical Outcomes through At-Home, Longitudinal Health Monitoring**
UbiComp/ISWC Doctoral Colloquium
[Richard Li](#)
- 2024 **Exploring and Characterizing Large Language Models for Embedded System Development and Debugging**
ACM CHI Late-Breaking Work
 Zachary Englhardt, [Richard Li](#), Dilini Nissanka, Zhihan Zhang, Girish Narayanswamy, Joseph Breda, Xin Liu, Shwetak Patel, Vikram Iyer
- 2024 **Opportunistic Nudges for Task Migration Between Personal Devices**
ACM CHI Late-Breaking Work
 Nikhita Joshi, [Richard Li](#), Jiannan Li, Leonardo Pavanatto, Michel Pahud, Jatin Sharma, Bongshin Lee, Hugo Romat, William Buxton, Nicolai Marquardt
- 2024 **The "Get Moving Trial": A Phase I/II RCT of Home-Based (P)rehabilitation ((P)REHAB) with ExerciseRx in Muscle-Invasive Bladder Cancer (MIBC)**
ASCO Journal of Clinical Oncology
 Hanna Hunter, Cindy Lin, [Richard Li](#), Otari Ioseliani, Leah Cantor, Elena Brewer, Samia Jannat, Karla Landis, David Bridges, Sean Munson
- 2023 **Smart-Pikachu: Extending Interactivity of Stuffed Animals with Large Language Models**
ACM UIST Student Innovation Challenge
 Toma Itagaki, [Richard Li](#)
- 2022 **Accuracy of Home Exercise Detection Using Novel Sonar Mobile Application Technology**
ACSM Medicine & Science in Sports & Exercise
 Ahmed I El-Sayed, [Richard Li](#), Parker Ruth, Libby Lavitt, Matthew Essman, Shwetak N Patel, Cindy Y Lin
- 2021 **Mobile, Hands-Free, Silent Speech Texting using SilentSpeller**
ACM CHI Late-Breaking Work
 Naoki Kimura, Tan Gemicioglu, Jonathan Womack, [Richard Li](#), Yuhui Zhao, Abdelkareem Bedri, Alex Olwal, Jun Rekimoto, Thad Starner

- 2020 **Evaluating a Novel, Portable, Self-Administered Device ("Flicker-App") that Measures Critical Flicker Frequency as a Test for Hepatic Encephalopathy in Patients with Cirrhosis**
AASLD The Liver Meeting Digital Experience
 Philip Vutien, Ravi Karkar, Richard Li, Kara Walter, Sean Munson, James Fogarty, George Ioannou
- 2016 **Tactile Teacher: Enhancing Traditional Piano Lessons with Tactile Instructions**
ACM Conference on Computer Supported Cooperative Work and Social Computing Poster
Richard Li, Yingyan Wang, Chih-Pin Hsiao, Nicholas Davis, James Hallam, Ellen Yi-Luen Do
- 2015 **Tactile Teacher: Sensing Finger Tapping in Piano Playing**
ACM Tangible, Embedded, and Embodied Interaction Demo
 Chih-Pin Hsiao, Richard Li, Xinyan Yan, Ellen Yi-Luen Do

Invited Talks

- 2025 **Designing and Deploying Next-Generation Health Technologies: Producing Actionable Insights from Novel Biomarkers**
 Host: Interactive Intelligent Systems Laboratory (University of Tokyo)
- 2025 **Designing and Deploying Next-Generation Health Technologies: Producing Actionable Insights from Novel Biomarkers**
 Host: Human-Computer Interaction Seminar (Columbia University)
- 2024 **Towards Personal Interpretation of Health Data for Improving Medical Outcomes**
 Host: VAK Sustainable Computing Lab (Northwestern University)
- 2024 **Supporting Exercise as a Medicine using Generative AI**
 Host: Computer Science & Engineering Industry Affiliates Day (UW)
- 2024 **Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency**
 Host: Advanced Topics in Human-Computer Interaction (UW CSE 510)
- 2024 **ExerciseRx-CP: At-Home Sensor-Driven Strengthening**
 Host: Cerebral Palsy Grand Rounds (Seattle Children's Hospital)
- 2023 **Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency**
 Host: Computer Science & Engineering Industry Affiliates Day (UW)
- 2022 **Sensing Device Formations using Ultra-Wideband Radios**
 Host: Human-Computer Interaction Seminar (Microsoft Research)
- 2022 **Critical Flicker Frequency for Minimal Hepatic Encephalopathy**
 Host: Hepatology Grand Rounds (Veterans Affairs Puget Sound Medical Center)

Teaching

- Fall 2024 **Embedded Systems Capstone (UW CSE/EE475, EE 542) - Instructor**
 Prepared and taught lectures on advanced embedded systems topics (i.e., PCB design, wireless communication, on-device machine learning). Guided 9 team projects from ideation to presentation. 36 students, overall rating of 4.5/5.
- Fall 2021 - Spring 2025 **Ubiquitous Computing Seminar (UW CSE 599U) - Instructor**
 Organized weekly seminar presentations on literature from ubiquitous computing, human-computer interaction, and mobile systems.
- Winter 2020 **Introduction to Human-Computer Interaction (UW CSE 440) - Teaching Assistant**
 Conducted lab sessions and graded assignments. 32 students.

Spring 2015 - Spring 2016	Technical Interview Prep (GT TIP) - Founder and Instructor Designed curriculum to prepare students for technical software engineering interviews, presented weekly lectures and organized class activities.
Spring 2015	Introduction to Computer Science (GT CS 1100) - Teaching Assistant Presented lectures on opportunities for computer science majors.

Student Mentorship

2024 - Present	Kanav Arora Deep learning for health sensing.
2022 - Present	Jason Miller Communication over audio, deep learning for pose tracking.
2022 - Present	Jonathan Shu - <i>Now: Software Engineer at Amazon.com</i> <u>UW Undergraduate Research Symposium</u> . Gaze tracking, Beacon hardware.
2021 - 2024	Toma Itagaki - <i>Now: PhD Student at Columbia University</i> <u>NSF GRFP</u> . Brain-computer interfaces, health sensing.
2020 - 2021	Millicent Li - <i>Now: PhD Student at Northeastern University</i> . <u>NSF GRFP</u> , <u>UW Undergraduate Honors Thesis</u> . Brain-computer interfaces.
2020 - 2023	Annalice Ni - <i>Now: Software Engineer at Bytedance</i> <u>UW Undergraduate Honors Thesis</u> . Fabrication and sensing techniques for interactive cardboard widgets.
2020 - 2023	Eddy Zhou - <i>Now: Software Engineer at F5</i> Fabrication and sensing techniques for interactive cardboard widgets.
2020 - 2023	Victor Chen - <i>Now: MS Student at Stanford, Software Engineer at Zoom</i> Sensing and interaction techniques for low-cost VR headsets.
2018 - 2019	Sophia Sun - <i>Now: Software Engineer at Yelp, PhD student at University of Minnesota</i> <u>GT Undergraduate Honors Thesis</u> . Face sensing for low-cost VR headsets.
2018 - 2019	Ruihan Xu - <i>Now: Software Engineer at Google</i> <u>GT Undergraduate Honors Thesis</u> . Face sensing for low-cost VR headsets.

Service

Paper Reviewing

Asterisk () denotes outstanding review*

ACM CHI - 2019, 2020, 2021, 2022, 2023, 2024*, 2025

ACM IMWUT - 2019, 2021*, 2022*, 2023, 2024

ACM ISWC - 2017, 2018, 2020

ACM UIST - 2021, 2022, 2025

ACM Health - 2025

NeurIPS TS4H - 2025

IEEE TVCG - 2024

ACM TEI - 2023

IEEE VR - 2022

ACM MobileHCI - 2022

IEEE ISMAR - 2020*

Student Volunteering

UbiComp - 2019

UIST - 2022

Institution Volunteering

UW Faculty Recruiting Host (HCI) - 2024

UW CSE Visit Days Area Organizer (UbiComp) - 2022 - 2024

UW CSE PhD Admissions Area Chair (HCI and UbiComp) - 2023

UW CSE PhD Admissions Reviewer - 2022 - 2024

UW CSE PhD Pre-Application Mentorship Service (PAMS) Mentor - 2021 - 2024